Department of Computer Science and Engineering
Shiv Nadar University Chennai
July-November 2026 Semester
M.Tech. Course on Deep Learning
Lab Assignment 4

Date: 1st September, 2026

1. **For 36-dimensional data of 5 image classes in Dataset2, build a multilayer feedforward neural network with two hidden layers with 30 nodes in the first hidden layer and 20 nodes in the second hidden layer. Use the softmax as the activation function in the output layer and the cross-entropy error as the error function. The stopping criterion is that the change in the average training error should be below a suitable threshold.**

   Configuration: ReLU function as the activation function in the hidden layers.

   Mode of learning: Mini-batch mode with the number of subsets as 5

   Parameter update rule: AdaM rule

   Study the performance of the model for each of the following methods of regularization:

   1. No regularization

   2. L_2-norm regularization with (a) $\lambda = 0.1$, (b) $\lambda = 1.0$ and (c) $\lambda = 10.0$

   3. Dropout with drop rate of (a) 10% and (b) 20%

   4. Dropconnect with drop rate of (a) 5% and (b) 10%

   5. Post-activation batch normalization

   The random initial weights should be the same for all the above studies.

   For each of the studies, give the following:

   1. Error vs Epoch plots for (i) the training data error and (ii) the validation data error.

   2. Classification accuracy and confusion matrix for (i) training data, (ii) validation data and (iii) test data.